

CenDox 20 Water Soluble Powder

FOR ANIMAL USE ONLY

Brand or Product Name :

CenDox 20 Water Soluble Powder

Name and Strength of Active Substance(s)

Each gram of CenDox 20 water soluble powder contains Doxycycline Hyclate 231 mg which is equivalent to Doxycycline 200mg .

Product Description

CenDox 20 water soluble powder is a light yellow fine powder packed in a 250g, 500g or 1000g aluminium sachet.

Pharmacodynamics Properties

Doxycycline is a semi-synthetic tetracyclines that is derived from oxytetracycline. Doxycycline hyclate may also be known as doxycycline hydrochloride. Doxycycline generally act as bacteriostatic antibiotics and inhibit protein synthesis by reversibly binding to 30S ribosomal subunits of susceptible organisms, thereby preventing binding to those ribosomes of aminoacyl transfer-RNA. Doxycycline also is believed to reversibly bind to 50S ribosomes and additionally alter cytoplasmic membrane permeability in susceptible organisms.

Pharmacokinetics Properties

Doxycycline is bio-available after oral administration. When orally administered, it reaches values greater than 70% in most species.

Feeding can modify the oral bioavailability of Doxycycline. In fasting conditions bioavailability is around 10 – 15% greater than when the animal is fed. Doxycycline is well distributed through the body as it is highly lipid soluble. It reaches well irrigated tissues as well as peripheral ones. It accumulates in liver, kidney, bones and intestine; enterohepatic recycling occurs. In lungs it always reaches higher concentrations than in plasma. Therapeutic concentrations have been detected in aqueous humour, myocardium, reproductive tissues, brain and mammary gland. Plasma protein binding is 90 – 92%. 40% of drug is metabolized and largely excreted through faeces (biliary and intestinal route), mainly as microbiologically inactive conjugates.

CHICKENS

After oral administration, doxycycline is quickly absorbed, achieving maximum concentrations (Cmax) around 1.5 h. Bioavailability is 75%. Absorption is decreased in the presence of feed in the gastrointestinal tract, bioavailability is then around 60% and the time to achieve the maximum concentration peak is significantly prolonged, (Tmax) 3.3 h.

Indication

Chickens (broilers) : Prevention and treatment of Chronic Respiratory Disease (CRD) caused by *Mycoplasma gallisepticum*.

Recommended Dosage

For chicken use :

20mg product / kg body weight = 100 mg of CenDox 20 per kg body weight per day for 3 - 5 days.

mg product / L water = [(100 mg CenDox 20 / kg body weight / day) x mean body weight (kg) of the animals to be treated] ÷ [water consumed by kg b.w (L)]

For the preparation of the medicated water the body weight of the animals to be treated and their actual daily water intake should be taken into due account.

To provide the required amount of active substance in mg per litre drinking water the following calculation should be made:

$$\frac{(100 \text{ mg CenDox 20 / kg bodyweight / day}) \times (\text{Average body weight (kg) of the animals to be treated})}{\text{Average amount of drinking water / animal (L)}} = \text{Amount (mg) of CenDox 20 / L of drinking water}$$

To ensure a correct dosage body weight should be determined as accurately as possible to avoid under dosing. The intake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of doxycycline has to be adjusted accordingly. Do not use at concentrations lower than 0.58 g of powder / l in drinking water with pH higher or equal to 7.5 to avoid precipitation. Sufficient access to the system of water supply should be available for the animals to be treated to ensure adequate water consumption. No other source of drinking water should be available during the medication period. Only sufficient medicated drinking water should be prepared to cover daily requirements. Medicated water should be replaced every 24 hour

Dosage schedule and duration of treatment : Once a day or divide dose into twice daily dosing (morning and evening) for 3 to 5 consecutive days.

Mode of Administration

To be administered via animal drinking water. Mixing Directions : To assure thorough dissolution, place the CenDox20 powder in a small mixing container and add 25 Litres of water . Stir until all the powder completely dissolved. Mix this concentrated stock solution with water to make the desired final volume of treated animal drinking water. Prepare fresh medicated solution daily.

Contraindications

Do not use in case of hypersensitivity to tetracyclines or to any of the excipients.

Do not administer to animals with severe liver- or kidney insufficiency.

Warnings and Precautions

Special precautions for use in animals

Eggs produced during medication should not be used for human consumption. The medicated water should be used during 24 hours. Due to likely variability in susceptibility of bacteria to doxycycline, bacteriological sampling and susceptibility testing are recommended. Under-dosing and/or treating for an insufficient length of time are considered to promote the development of resistance in bacteria and should be avoided. Avoid administration in oxidized drinking equipment. Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local (regional and farm level) epidemiological information about susceptibility of the target bacteria as well as by taking into account official national antimicrobial policies. Inappropriate use of the product may increase the prevalence of bacteria resistant to doxycycline and may decrease the effectiveness of treatment with other tetracyclines due to the potential for cross-resistance.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

If you know you are allergic to the tetracycline class of antibiotics, special care should be taken when handling this product or the medicated solution. During preparation and administration of the medicated drinking water, skin contact with the product and inhalation of dust particles should be avoided. Wear impermeable gloves and an appropriate dust mask when applying the product. In the event of eye or skin contact, rinse the affected area with large amounts of clean water and if irritation occurs, seek medical attention. Wash hands and contaminated skin immediately after handling the product. If you develop symptoms following exposure such as skin rash, you should seek medical advice and show this warning to the physician. Swelling of the face, lips or eyes, or difficulty with breathing are more serious symptoms and require urgent medical attention. Do not smoke, eat or drink while handling the product. Take measures to avoid producing dust when incorporating the product into water. Avoid direct contact with skin and eyes when handling the product to prevent sensitization and contact dermatitis.

Interactions with Other Medicaments

Do not use in conjunction with bactericidal antibiotics, such as penicillins and cephalosporins.

Tetracyclines can chelate cations (e.g. Mg, Mn, Fe and Al) and this may lead to decreased bioavailability

Usage during pregnancy and lactation

Do not use in laying birds producing eggs for human consumption.

Adverse Effects

None known.

Overdose and Treatment

The administration of 80mg/kg in chickens (corresponding to 4 times the recommended dose), for 5 days did not cause any adverse reaction. In case of overdose the treatment should be suspended and symptomatic treatment established.

Withdrawal Period(s)

Chickens: 6 days

Not authorized for use in laying birds producing eggs for human consumption.

Do no use within 4 weeks of onset of the laying period.

Shelf-life

Shelf life as packaged for sale: 36 months.

Shelf life after first opening of the immediate packaging: 24 hours

Shelf life after reconstitution in drinking water: 24 hours

Maximum Residual Limit (MRL)

Doxycycline : Muscle (poultry) is 100ug/kg, liver (poultry) is 300ug/kg, fat (poultry) is 300ug/kg and kidney (poultry) is 600ug/kg.

[Ref. REGOVP, Version 2, dated Dec 2009]

Storage Conditions

Store in a dry place below 25°C. Keep out of reach of children. Jauhi dari kanak-kanak. Any opened sachet or unused medicated water to be discarded after 24 hours.

Disposal of containers

Traces of medicated powder left in an opened aluminum sachet can be rinsed out with clean animal drinking water and use in animal treatment. Cleaned aluminum sachet can be safely disposed according to local government regulations.

Dosage Forms and packaging available

Available in 250g , 500g and 1000g powder per sachet which are packed in 10, 20, 50 and 100 sachets per box.

Name and Address of manufacturer

Life Biopharma Sdn. Bhd. (company no. 546163-D)

No. 250-251, Jalan Industri Galla 13, Kawasan Perindustrian Galla, 70200 Seremban, Negeri Sembilan, Malaysia.

Date of Revision of Package Insert

2013-07-29